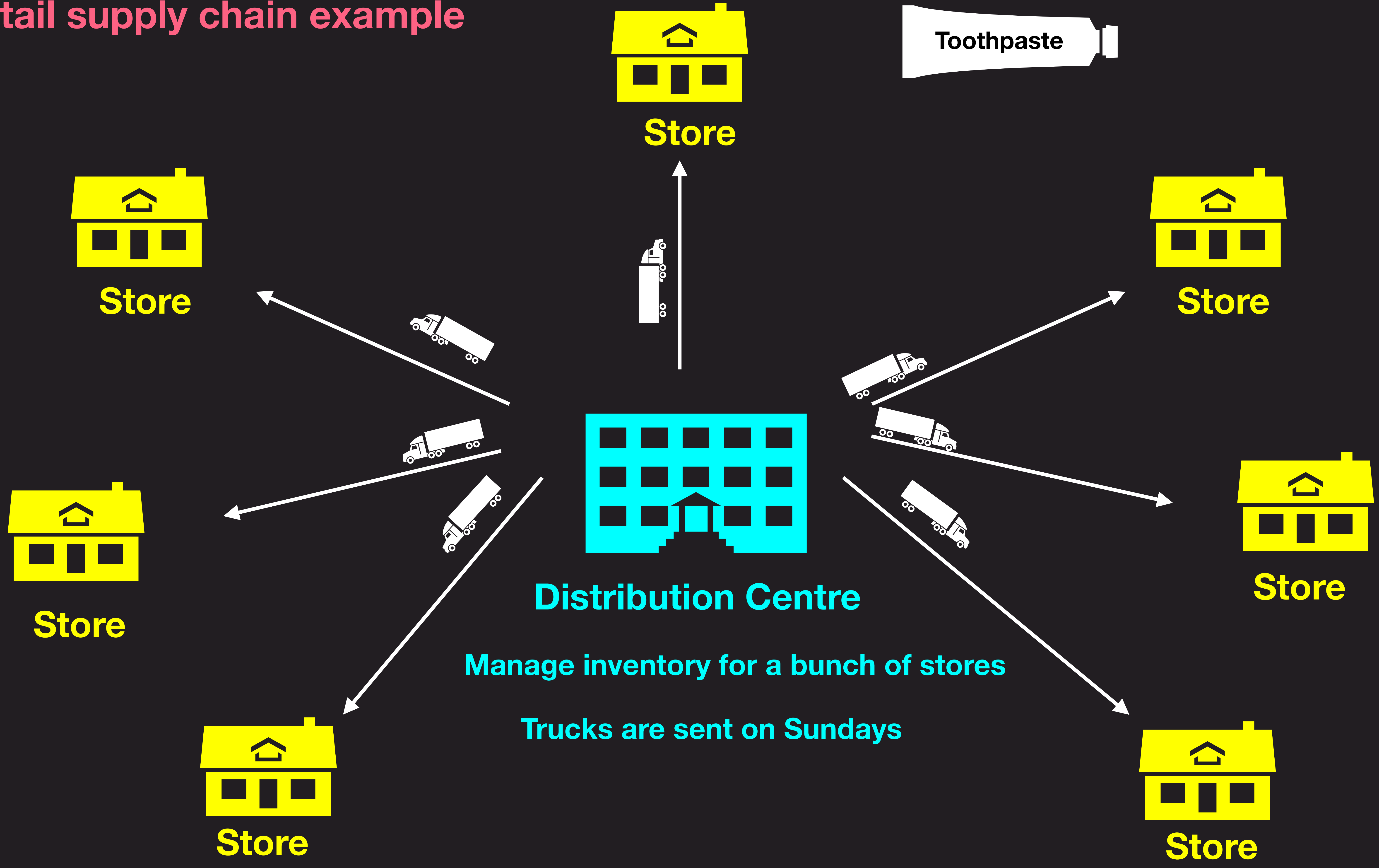
Distribution	Parameters	Equation	Mean	Variance	Code
Bernoulli	p	$P[X=1]=p$ $P[X=0]=1-p$	p	$p(1-p)$	bernoulli.pmf()
Binomial	n, p	$P[X=k] = {}^nC_k p^k (1-p)^{n-k}$	np	$np(1-p)$	binom.pmf()
Geometric	p	$P[X=k] = (1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	geom.pmf()
Poisson	λ	$P[X=k] = \frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	poisson.pmf()
Exponential	λ	$F(x) = 1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	expon.cdf()
Normal	μ, σ	$f(x) = \frac{1}{2\sigma^2} e^{\frac{(x-\mu)^2}{\sigma^2}}$	μ	σ	norm.cdf()
Log normal	μ, σ	$\log X \sim$ Gaussian	$exp(\mu + \sigma^2/2)$	$e^{(\sigma^2)-1} e^{2\mu+\sigma^2}$	lognorm.cdf()

Retail supply chain example



Pooled Blood test

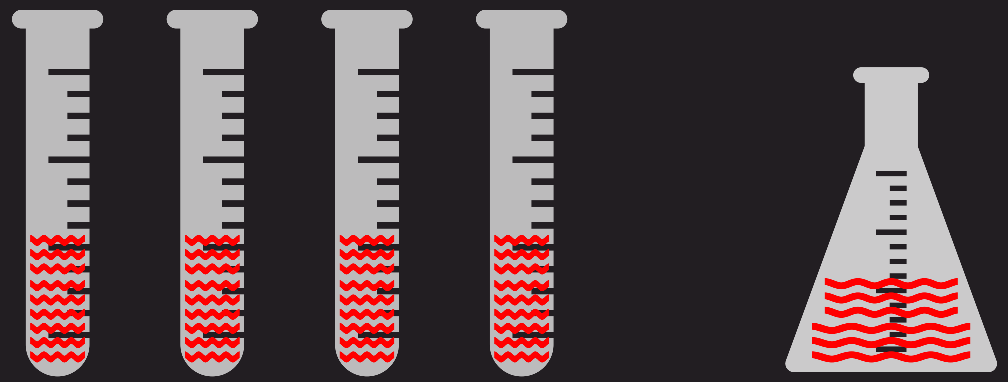
A blood bank tests pooled samples of 4 people at a time.

If clean, the bank stores all four.

If unacceptable, then all 4 samples are tested individually.

The probability of any sample being unacceptable is 0.1.

Find the expected number of tests needed



Probability of individual begin unacceptable: $p = 0.1$

Let X denote the number of tests needed

What is the question asking? $E[X]$

What are the values that X can take? 1 or 5

$P[X=1] =$ `binom.pmf(k=0, n=4, p=0.1)` $= 0.6561$

$$P[X=1] = {}^4C_0(0.1)^0(0.9)^4 = (0.9)^4 = 0.6561$$

$$P[X=5] = 1 - 0.6561 = 0.3439$$

$$\begin{aligned} E[X] &= 1 * P[X=1] + 5 * P[X=5] \\ &= 1 * 0.6561 + 5 * 0.3439 \\ &= 2.37 \end{aligned}$$

Battery → lifetime mean lifetime = 5 weeks.

std. deviation = 1.5 weeks.

Approx. the prob. of needing 13 or more batteries in one year.

$X_i \rightarrow$ lifetime of the i^{th} battery

$$Y = X_1 + X_2 + \cdots + X_{12}$$

$$EX_i = 5 \longrightarrow EY = 60$$

$$\mathrm{Var}\, X_i = 1.5^2 \rightarrow \mathrm{Var}\, Y = 12(1.5)^2$$

$$P[Y < 52] = P\left[\frac{Y-60}{\sqrt{12(1.5)^2}} < \underbrace{\frac{52-60}{\sqrt{12(1.5)^2}}}_{y_1}\right]$$

norm.cdf (y_1)

Airline Overbooking

Five percent of the people making reservations on a flight will not show up.
Suppose the airline sells 52 tickets for a flight that can hold only 50 passengers.
What is the probability that there will be a seat available for every passenger who shows up?

Probability of showing up: $p = 0.95$

Let X denote the number of people who show up

What are we asked to find? $P[X \leq 50]$

Can we convert this to asking this question:
A coin is tossed 52 times, the probability of heads is 0.95.
What is the probability of 50 or lesser heads?

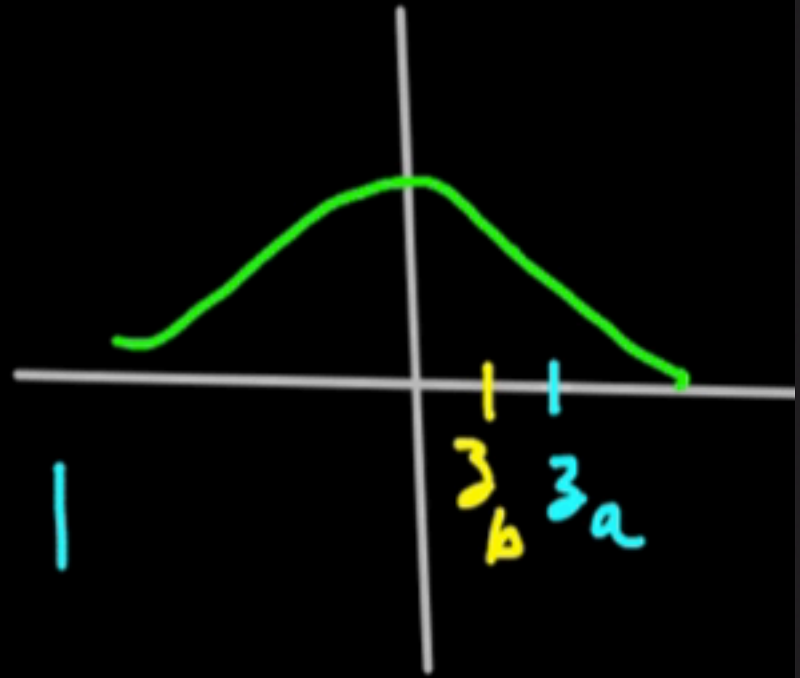
$P[X \leq 50] =$ `binom.cdf(k=50, n=52, p=0.95)` $= 0.74$

Two types of Exams - Range of marks

different range

$$\boxed{z = \frac{x-\mu}{\sigma}}$$

	Test 1	Test 2
mean	1500	21
std dev	300	5



A: test 1 $\longrightarrow$ 1800 $\longrightarrow$ $z_a = \dfrac{1800-1500}{300} = 1$

B: test 2 $\longrightarrow$ 24 $\longrightarrow$ $z_b = \dfrac{24-21}{5} = \dfrac{3}{5}$

z-score

Hiring

IQ is known to be approximately Gaussian with a mean of 100 and standard deviation of 15

You want to hire a person with IQ greater than 110

Conducting an interview costs 1000. What is the expected budget?

$$P[X > 110] = P\left[Z > \frac{(110 - 100)}{15}\right] = \text{1 - norm.cdf(10/15)} = 0.252$$

Let N denote the number of interviews till first hire

Which distribution does N follow? Geometric

What is $E[N]$? 1/0.252 = 3.96

Expected budget = 3960

Simulate a fair coin from a biased coin

There is a coin that lands heads 70% of the times

How can we use this coin so that it lands heads 50% of the times?

Verify the output of your algorithm using 10000 simulations at 95% confidence

Let us toss the biased coin twice

Sample space S = {HH, HT, TH, TT}

$$P[HH] = 0.7 * 0.7$$
$$P[HT] = 0.7 * 0.3$$
$$P[TH] = 0.3 * 0.7$$
$$P[TT] = 0.3 * 0.3$$

These two have same probability

Biased	Fair
$HH \rightarrow$	Ignore
$TT \rightarrow$	Ignore
$HT \rightarrow$	H
$TH \rightarrow$	T